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Lecture: Computer System

GROUP 6

Topic : Motion-Activated Electronic Notification System

Subject lecturer : Phạm Ngọc Thành

Name	Student ID	Role
Phạm Hải Nam	23071311	Leader
Nguyễn Bảo Khánh	23070228	Member

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Finally, we express our sincere appreciation to the Faculty and the University. We are grateful for the exceptional academic environment and the comprehensive resources provided, which served as the essential foundation for our research and the successful execution of this project.

ASSIGN TASKS

Member	Mission
Nguyen Bao Khanh – 23070228	Design and simulation of Configuration 1 (IR door-status chime system) in Proteus; CNY70/LM358B comparator front-end and W103 threshold calibration; firmware for door-status detection and buzzer/LED control; component sourcing, functional testing and report writing.
Phạm Hải Nam - 23071311	System architecture and lead coordination; design and simulation of Configuration 2 (PIR thermal sensing system) in Proteus; HC-SR501, C945 transistor and relay driver implementation; safe high-power load management; firmware for motion detection; testing, result analysis and report writing.

I. Introduction

1.1 Background And Concept

- In recent years, the rapid advancement of embedded systems and the Internet of Things (IoT) has significantly transformed residential automation and home security solutions. Among various smart home functionalities, automated entry detection and signaling play a critical role in modern residential management, structural surveillance, and institutional access control.

- Effective and immediate detection of human presence at entry points is essential for enhancing property security, improving visitor convenience, and supporting automated monitoring infrastructures. Although conventional security installations, such as continuous video surveillance or manual push-button doorbells, are widely deployed, they present distinct limitations, including high power consumption, reliance on manual visitor interaction, and a lack of localized feedback at the control node.

- The emergence of affordable yet highly capable microcontrollers, such as the ESP32 platform, integrated with advanced peripheral sensors and versatile communication interfaces, has opened up new possibilities for building compact, autonomous, and energy-efficient notification systems. Beyond basic hardware triggering, these microcontrollers enable real-time state processing and seamless integration with isolated higher-power alerting hardware.

- Motivated by these advantages, this project focuses on designing and implementing a Motion-Activated Electronic System utilizing an ESP32 microcontroller, a Passive Infrared (PIR) motion sensor capable of detecting human presence through body heat, and an integrated signaling interface. By combining hardware components, digital input/output mapping, and embedded programming into a complete working

prototype, the project aims to bridge the gap between theoretical computer logic knowledge and practical electronic engineering applications.

1.2 Problem Statement

- Traditional visitor notification systems are largely dependent on manual operation, requiring an individual to physically press a button to trigger an audible alert. This approach is frequently inefficient in scenarios involving delivery automation, hands-free entry requirements, or proactive perimeter security monitoring where detecting an approaching individual before they reach the door is paramount.
- On the other hand, many elementary motion-detection designs rely on basic continuous polling architectures without proper isolation mechanisms. This often results in system instability, high susceptibility to false triggers from power fluctuations, and potential electrical feedback hazards when high-voltage or high-current alerting devices are directly connected to sensitive microcontroller pins. Therefore, there is a distinct technical requirement for an automated notification framework that achieves an optimal balance between responsive detection accuracy and operational safety.
- The central engineering problem addressed in this project can be formulated as follows: *How can a reliable, highly responsive, and electrically isolated motion-activated alert system be designed and built using an ESP32 microcontroller and a PIR sensor, while ensuring accurate signal state processing and safe load management?*
- To resolve this challenge, the project strategically employs a PIR sensor as the primary detection element, the ESP32 as the central processing unit, and an optically/mechanically isolated relay module to manage the high-current buzzer loop safely.

1.3 Project Question

- This project is guided by a single central engineering question, supported by three sub-questions that the design and simulation set out to answer.
- **Main question:** *Can a low-cost, ESP32-based alert system detect an entry or presence event automatically and drive an isolated, higher-power alerting load safely and reliably, at a unit cost low enough to be commercially viable?*
- To answer this, the project addresses three sub-questions:
 - + **Detection:** Can an IR transmit/receive pair and an HC-SR501 PIR sensor, each conditioned by an LM358B stage, reliably convert a real-world event (a door opening, a person approaching) into a clean digital signal for the ESP32?
 - + **Safe actuation:** Can the system switch a high-power 12V load through a relay while keeping the 3.3V logic completely isolated and protected?
 - + **Feasibility:** Can the complete design be validated before manufacturing, and built at a competitive cost, so that it is attractive to a prospective customer or investor?

The remainder of this report answers these questions through the hardware design, the simulation and firmware, and the experimental results.

1.4 Project Objectives

- The overall objective is to design, simulate, and present a commercially viable Motion-Activated Electronic Alert System. The product is delivered as two independent configurations, each targeting a distinct detection principle and use case. The specific objectives of each configuration are stated below.
- **Configuration 1: Automated Door-Status Entry Chime System (Circuit 1):**

- + Establish a door-status monitoring baseline at the storefront entrance using a dedicated pair of infrared (IR) transmitting and receiving LEDs to detect the physical opening and closing actions of the door.
- + Digitize the variable analog signal from the IR receiver into stable binary logic states using the LM358B operational amplifier configured as a voltage comparator, mapping the alignment of the IR beam to the door's position.
- + Integrate a W103 10K potentiometer (trimmer) into the comparator circuit to allow manual tuning and calibration of the infrared detection threshold based on door clearance and ambient light.
- + Capture these digital state transitions through the GPIO interface of the ESP32 and trigger an automated welcome-chime cycle immediately upon detecting a door-opening event.
- + Safely drive a low-voltage signaling buzzer or melody module through a mechanical relay isolation framework, producing a welcoming chime immediately on each door-opening event.

- Configuration 2: Pyroelectric (PIR) Thermal Sensing Notification System (Circuit 2):

- + To detect human presence and movement within a localized wide-area zone by capturing passive body-heat radiation using the HC-SR501 PIR sensor.
- + To process the discrete high/low output signals generated by the HC-SR501 directly through the embedded programming loop of the ESP32.
- + To safely execute high-power load management by using a 5V mechanical relay module to trigger and switch an isolated 12V direct-current (DC) doorbell/alerting-lamp circuit.

- Common Project Objectives: To validate both configurations entirely within Proteus before committing to a physical prototype, thereby

lowering development cost and risk, and to deliver a project plan, bill of materials, and cost analysis that demonstrate commercial feasibility to a prospective investor or business partner.

1.5 Scope Of The Project

- Two independent single-node units, each built around one ESP32 board powered by an HW-131 supply module. Each unit functions as a complete, standalone product, allowing a customer to deploy either configuration on its own or both together depending on the needs of the premises.
- **Configuration 1:** IR transmit/receive LED pair, an LM358B comparator front-end, a W103 10K trimmer, a relay, and a buzzer/melody module. Together these implement automatic door-status sensing, threshold calibration, and a welcome-chime cycle at the storefront entrance.
- **Configuration 2:** HC-SR501 PIR sensor, an LM358B stage with a W103 10K trimmer, a C945 transistor relay driver, a 5V relay, and an isolated 12V alerting lamp/bell circuit. These provide wide-area human-presence detection and safe switching of a higher-power alerting load, suitable for both occupied-hours monitoring and after-hours security.
- Full schematic design and functional simulation of both circuits in Proteus, including the operating logic of the firmware. The simulation exercises every signal path, covering sensor input, LM358B conditioning, ESP32 decision-making, and the relay-driven outputs, so that the complete behaviour of each configuration is verified before any physical hardware is built, keeping development cost and risk low.

1.6 Target Customers And Stakeholders

- The product is designed for a broad range of buyers who share a need for affordable, automatic alerting but differ in how they use it:

+ **Convenience stores and retail shops:** the main buyers of Configuration 1. These businesses want an automatic entry chime that announces each customer the moment the door opens, without relying on a staff member to watch the entrance or a visitor to press a button. For high-traffic stores, this improves customer service and ensures no one enters unnoticed.

+ **Warehouses, offices, and property owners:** the main buyers of Configuration 2. These customers need wide-area presence detection to monitor larger spaces and entry points, particularly for after-hours security where an immediate, conspicuous 12V alert deters intruders and notifies staff of any movement on the premises.

+ **Hospitality venues and clinics:** reception areas such as hotels, restaurants, and medical offices benefit from a discreet automatic visitor tone that signals an arriving guest or patient, allowing front-desk staff to respond promptly even when attending to other tasks.

+ **System integrators and electronics distributors:** rather than end users, these are commercial partners who may resell the device or bundle it into larger security and store-automation packages. Their interest lies in the product's low unit cost and its flexible, shared-platform design, which makes it easy to integrate into existing offerings.

- **Other Stakeholders :** Beyond the direct buyers, several other parties have a stake in the success of the product and were considered throughout its design:

+ **Investors / business partners:** those who fund the move from a validated prototype to full production. They expect a clear cost breakdown and a convincing feasibility case before committing capital, which is why

this proposal emphasizes the simulation evidence, the low bill of materials, and a realistic production roadmap.

+ **Manufacturers:** the parties responsible for building the product at scale. They benefit directly from the shared-platform design, which reuses a single ESP32 core and one firmware base across both configurations. This lowers the bill of materials, simplifies inventory and assembly, and allows either product, or a combined unit, to be built from one common set of components.

+ **End users:** the staff and security personnel who interact with the alert every day. For them, the system must be immediate in its response, reliable over long periods of continuous operation, and simple enough to understand and trust without technical training.

1.7 Novelty And Contribution

- While the individual components used in this project are standard, the novelty lies in how they are combined and packaged into a commercially-oriented product. The specific novel contributions of each project component are:

+ **Dual-configuration shared platform (system-level novelty):** unlike typical single-purpose alert circuits, this project delivers two distinct products, door-status chiming and wide-area presence alerting, from a single shared ESP32 core, power module, conditioning stage, and firmware base. This shared-platform approach is the central novel idea, lowering cost and inventory for a manufacturer.

+ **Hardware-tunable detection without firmware changes (Configuration 1):** the IR front-end uses a W103 trimmer with the LM358B comparator so the detection threshold can be re-calibrated in hardware for different door clearances and lighting, without reprogramming, a practical novelty for field installation.

+ **Safe transistor-relay isolation stage (Configuration 2):** the C945 transistor driver plus relay forms a clearly partitioned crossing point

between the 3.3V logic and an isolated 12V load, designed so that a fault on the high-power side can never reach the microcontroller, a contribution over simpler direct-drive motion alarms.

+ **Simulation-first validation as a risk-reduction method:** by validating both configurations in Proteus with the same firmware binary that runs on the physical chip, the project demonstrates a low-risk path from design to product, a methodological contribution that supports the feasibility argument.

- Together these contributions turn a set of standard components into a novel, low-cost, two-in-one alert product with a built-in upgrade path toward connected (Wi-Fi) operation.

II. System Overview

2.1 Overall System Description

- The Motion-Activated Electronic Alert System is delivered as two independent configurations that share a common design philosophy: a sensing layer observes the environment, a processing layer built on the ESP32 interprets the signals and applies decision logic, and an output layer produces the alert and switches any external load. Each configuration is a complete, standalone product captured as its own Proteus schematic; a customer may deploy either one, or both, depending on whether they need an automatic entry chime, after-hours presence alerting, or both.

- Configuration 1 (Circuit 1) is built around an infrared transmit/receive LED pair aimed across the doorway. While the door is closed the beam relationship is constant; opening the door changes the reflected or interrupted IR level. The LM358B comparator, with its threshold set by a W103 10K trimmer, converts this analog change into a clean digital edge that the ESP32 reads. On a door-opening event the ESP32 sounds a chime

through a relay-isolated buzzer module. Power is supplied by an HW-131 module.

- Configuration 2 (Circuit 2) is built around an HC-SR501 PIR sensor that detects the body-heat radiation of a person moving in its wide-area field of view. Its digital output is processed directly by the ESP32 firmware, which then drives a C945 transistor that switches the 5V relay coil to energize an isolated 12V DC alerting lamp/bell. An LM358B stage with a W103 10K trimmer and the HW-131 supply complete the circuit.
- Treating the two configurations as separate products is also a commercial advantage: a manufacturer can sell a low-cost entry-chime unit, a presence-alert unit, or both, while reusing the same ESP32 processing core and firmware framework.

2.2 Input – Processing – Output (IPO) Model

- The behaviour of each configuration is described precisely by the Input-Processing-Output model:

Configuration 1 - IR Door-Status Chime:

- + **Input:** A customer opens the storefront door. This changes the amount of infrared light seen by the IR receiver LED relative to the transmitter, altering the receiver's analog output voltage.
- + **Conditioning:** The LM358B comparator compares the receiver voltage against the reference set by the W103 10K trimmer, producing a clean digital edge that marks the transition between door-closed and door-open.
- + **Processing:** The ESP32 reads the digital edge on a GPIO input, recognizes a door-opening event, and starts the welcome-chime cycle.
- + **Output:** The ESP32 drives the buzzer/melody module through a relay, which isolates the signaling circuit from the logic, to sound the welcome chime on each door-opening event.

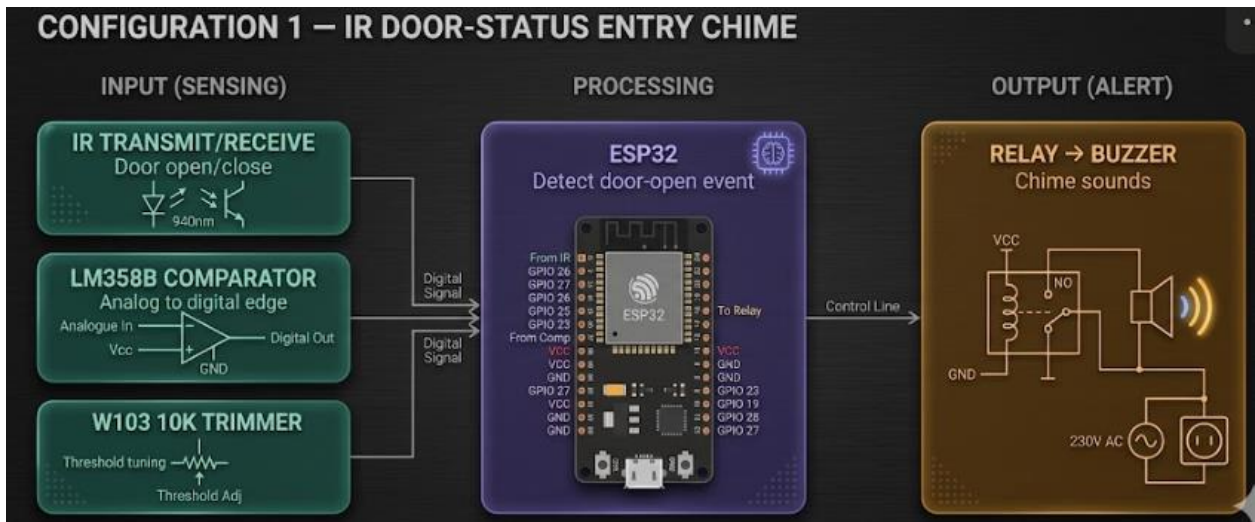


Figure 1: IR-Based Door Monitoring System Architecture

Configuration 2 - PIR Thermal Notification:

- + **Input:** A person enters the monitored zone; their body-heat radiation moving across the sensor's field of view is captured by the HC-SR501.
- + **Processing:** The HC-SR501 outputs a digital HIGH on motion; the ESP32 reads this directly in its programming loop and decides to raise the alert.
- + **Output:** The ESP32 turns on the C945 transistor, which drives the 5V relay coil; the relay contacts switch the isolated 12V DC circuit to light the alerting lamp or sound the bell, keeping the high-power load fully separated from the logic.

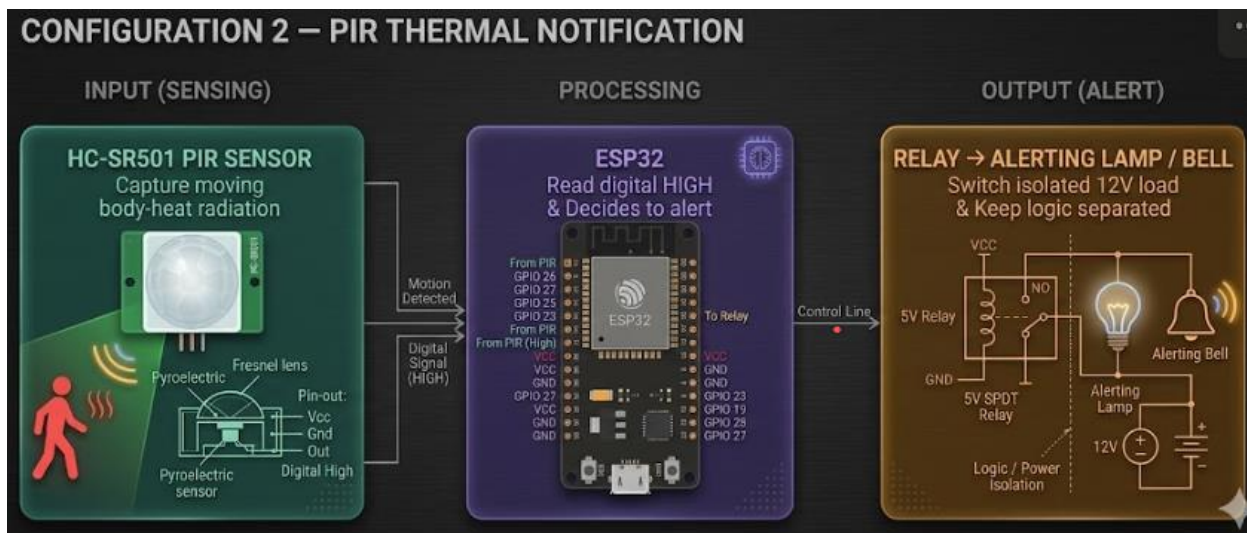


Figure 2: PIR-Based Motion Detection System Architecture

2.3 Role of Each System Component

- Configuration 1 — Automated Door-Status Entry Chime System:

+ **ESP32:** Acts as the central processing unit. It reads the comparator input on its GPIO, detects the door-opening event, and sequences the chime output.

+ **HW-131 power module:** A breadboard power-supply adapter that provides regulated 3.3V and 5V rails to the ESP32 and the surrounding circuitry.

+ **IR transmit/receive LED pair:** The door-status sensor. The transmitter emits infrared light and the receiver detects the change caused by the door opening or closing, producing the raw analog signal.

+ **LM358B op-amp:** Configured as a voltage comparator that digitizes the weak IR-receiver signal into a clean, stable logic level for the ESP32.

+ **W103 10K potentiometer (trimmer):** Sets the comparator's reference voltage, allowing the detection threshold to be tuned to the door clearance and ambient light.

+ **Buzzer / melody module:** Plays the welcome chime (Circle K-style notification) on each door-opening event.

- Configuration 2 — Pyroelectric (PIR) Thermal Sensing Notification System:

- + **ESP32:** Acts as the central processing unit. It reads the HC-SR501 output on its GPIO, confirms the motion event, and drives the relay control pin through the C945 transistor.
- + **HW-131 power module:** Provides the regulated 3.3V and 5V rails to the ESP32 and the surrounding circuitry.
- + **HC-SR501 PIR sensor:** Detects human presence and movement by capturing passive body-heat radiation across a wide area.
- + **LM358B op-amp:** Conditions and buffers the sensor signal where required.
- + **W103 10K potentiometer (trimmer):** Sets the reference/sensitivity for the LM358B stage, allowing the motion-detection threshold to be tuned during installation.
- + **C945 transistor:** An NPN transistor that acts as a switching driver between the ESP32 and the relay coil. The ESP32's low-current GPIO pin cannot drive the relay directly, so the C945 amplifies the control signal to supply the coil current needed to energize the relay.
- + **Relay (5V):** An electromechanical switch that isolates the low-voltage logic from the high-power 12V load circuit.
- + **12V alerting lamp / bell:** The high-power output load switched by the relay to signal detected presence.

→ **Overall:** The two configurations demonstrate how a single common platform can serve two distinct purposes by changing mainly the sensor and the output device. In both cases the ESP32 is the central decision-maker, the HW-131 module supplies regulated power, the LM358B handles the sensor signal, and the relay provides electrical isolation between the low-voltage logic and the output load. Configuration 1 specializes this platform for door-status sensing with an IR LED pair and a buzzer chime, while Configuration 2 specializes it for wide-area

presence detection with an HC-SR501 sensor driving a 12V lamp/bell through a C945 transistor and relay. This shared-core, swappable-periphery approach keeps the design simple, lowers the bill of materials, and lets a manufacturer build either product, or a combined unit, from one common set of parts and a single firmware base.

III. Hardware Architecture

3.1 ESP32 Board Architecture

- The ESP32-WROOM is a low-cost, low-power system-on-chip module that integrates processing, memory, wireless connectivity, and a rich set of peripheral interfaces on a single compact board. In both configurations of this project, the ESP32 serves as the main hardware component responsible for sensor data acquisition, signal processing, decision logic, and output control. Its small form factor combined with substantial computational power makes it well-suited for the embedded alert applications developed here.

- The ESP32 was developed by Espressif Systems, a semiconductor company based in Shanghai, China, and was first released in 2016 as the successor to the earlier and highly popular ESP8266. While the ESP8266 had already gained wide adoption among hobbyists and industry for its affordable Wi-Fi capability, it was limited by a single-core processor and a modest set of peripherals. The ESP32 was designed to overcome these limitations by introducing a dual-core processor, integrated Bluetooth alongside Wi-Fi, a richer set of peripheral interfaces, and improved power management. Owing to this combination of capability and low cost, the ESP32 quickly became one of the most widely used platforms in embedded systems, the Internet of Things (IoT), and academic projects worldwide. The ESP32-WROOM module used in this project is one of the most common packaged forms of the chip, integrating the SoC, flash memory, and antenna into a single ready-to-use unit.

- The internal architecture of the ESP32 can be divided into several key subsystems, including the processor, memory, general-purpose input/output (GPIO), and communication interfaces. These subsystems work together to support interaction with the external sensors and output devices of each circuit.



Figure 3: ESP32

a. Processor (CPU):

- At the core of the module is the Espressif ESP32 SoC, which features a dual-core Tensilica Xtensa LX6 processor running at up to 240 MHz. This processor provides more than sufficient performance to execute the firmware, run the detection state machine, and handle real-time signal acquisition from the IR comparator or the HC-SR501 sensor without any noticeable latency.

b. Memory:

- The module integrates 520 KB of internal SRAM together with typically 4 MB of external flash memory. The flash holds the firmware and configuration data, while the SRAM holds the runtime variables and detection state. This capacity is more than adequate for the alert logic and

leaves substantial headroom for future features such as event logging or wireless connectivity.

c. General-Purpose Input/Output (GPIO):

- The board provides a large number of multifunction GPIO pins operating at 3.3 V logic level. These pins support digital input/output, analog-to-digital conversion, PWM, and the I²C, SPI, and UART buses. In Configuration 1, an input pin reads the LM358B comparator output while output pins drive the relay and buzzer. In Configuration 2, an input pin reads the HC-SR501 output and an output pin drives the C945 transistor that controls the relay coil.

d. Communication Interfaces:

- The ESP32 supports several communication protocols essential for embedded applications: I²C, SPI, and UART for peripheral connections, together with integrated Wi-Fi 802.11 b/g/n and Bluetooth/BLE radios. None of the wireless functions are used in this baseline design, but their presence enables straightforward future expansion toward remote monitoring or IoT integration with no change to the sensing or output hardware.

e. Power Supply:

- The ESP32 operates from a regulated 5 V input through its on-board regulator, which derives the 3.3 V rail used by the core logic. The board can also supply regulated 3.3 V and 5 V to external low-power peripherals such as the sensors and the op-amp used in this project.

→ **Overall:** The architecture of the ESP32 provides a balanced combination of processing capability, flexible I/O, isolation-friendly output pins, and rich communication interfaces, making it a suitable platform for implementing both configurations of the proposed alert system.

3.2 ESP32 as the Central Processing Unit

- In both configurations the ESP32 acts as the central processing and control unit, coordinating all hardware operations. It is responsible for reading the conditioned sensor signals, applying the detection and debounce logic, and controlling the output stage within a unified firmware architecture.
- The ESP32 polls its digital input pins, timestamps each state change, and applies the decision rule that distinguishes a genuine event from noise. By controlling the timing of detection, the alert duration, and the cooldown interval, it ensures synchronized operation between the sensing, processing, and output stages.
- Running its firmware directly on the SoC, the ESP32 enables a close interaction between hardware and software. Because all logic resides in firmware, the system behaviour (threshold response, chime or alert duration, debounce window) can be reconfigured without any modification to the underlying hardware design, which is a significant advantage for a manufacturer maintaining several product tiers.

3.3 Power Supply Module (HW-131)

- The HW-131 is a breadboard power-supply module used to deliver stable, regulated voltage to both configurations. It accepts an external DC input and provides selectable regulated 3.3 V and 5 V output rails, supplying the ESP32 and the surrounding circuitry (the op-amp, sensors, and relay coil drive). By providing clean, properly matched voltage rails, the HW-131 minimizes the risk of malfunction, false triggers from voltage fluctuations, and communication errors during prolonged operation. Its dual-rail output is particularly convenient here, as the logic devices operate at 3.3 V/5 V while still allowing the relay and its driver to be supplied at the level they require.

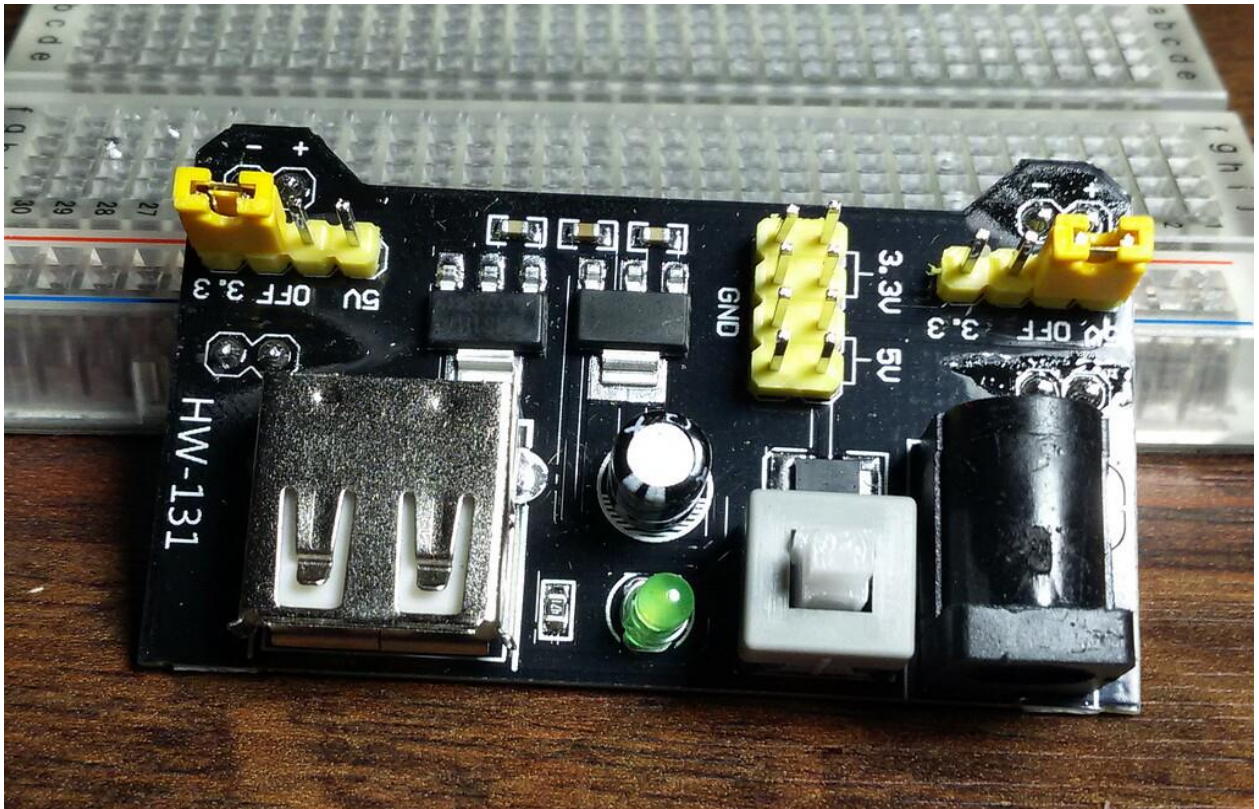


Figure 4: HW-131

3.4 Breadboard

- The breadboard is a solderless prototyping board used to assemble and interconnect the components of both configurations. Its surface consists of many contact holes joined internally by conductive rows and columns: the power rails running along each side distribute the supply voltage and ground, while the rows in the central area connect the component leads to one another. This allows the ESP32, the HW-131 module, the LM358B op-amp, the W103 trimmer, the sensors, the C945 transistor, and the relay to be inserted directly and wired together with jumper wires, without any soldering.

- In this project, the breadboard acts as the central assembly base: the HW-131 module feeds the 3.3 V and 5 V rails into the board's power rails, from which every component draws its supply and shares a common ground. Using a breadboard offers two important benefits during development. First, it allows rapid modification and testing, since components can be repositioned, alternative values tried, or the W103 trimmer recalibrated

easily without damaging any parts. Second, it keeps the circuit tidy and organized, reducing the likelihood of wiring errors and making inspection and debugging convenient.

- It should be noted that the breadboard is a tool for the prototyping and simulation stage only. When the product moves to production, the temporary breadboard connections would be replaced by a purpose-designed printed circuit board (PCB) to improve mechanical durability, electrical stability, and suitability for mass manufacturing.

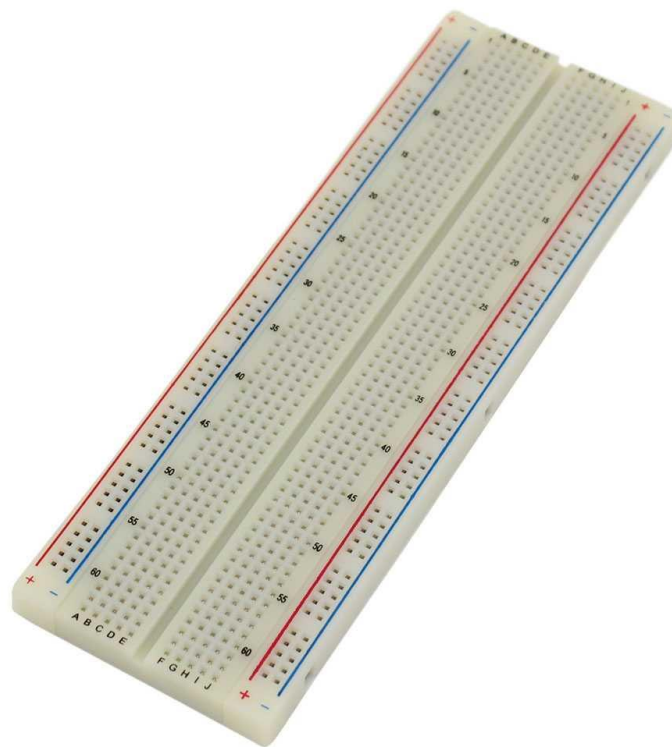


Figure 4: Breadboard

3.5 Infrared Transmit/Receive LED Pair (Configuration 1)

- The infrared transmit/receive LED pair is the primary sensing element of Configuration 1. It consists of an infrared emitting LED (the transmitter) that continuously radiates infrared light, and an infrared receiver positioned so that opening or closing the door changes the amount of infrared light reaching it. This change produces a variable analog voltage at the receiver node.

- Unlike a digital sensor, the IR pair produces a weak, continuously varying analog signal rather than a clean logic level. For this reason it is paired with the LM358B comparator stage, which digitizes the signal. The detection distance and reliability depend on the alignment of the transmitter and receiver and on the surrounding ambient light, which is why a calibration element (the W103 trimmer) is included in the conditioning circuit.



Figure 6: Transmit/Receive LED Pair

3.6 LM358B Operational Amplifier

- The LM358B is a dual low-power operational amplifier used for signal conditioning in both configurations. In Configuration 1 it is configured as a voltage comparator: the analog signal from the IR receiver is applied to one input, while the other input is held at a reference voltage set by the W103 trimmer. When the receiver voltage crosses the reference, the comparator output snaps sharply between its low and high states, converting the gradual analog change into a clean, stable digital edge that the ESP32 can read directly.

- In Configuration 2 the LM358B serves as a conditioning and buffering stage for the sensor signal where required, ensuring the level presented to the ESP32 is well-defined. Using the op-amp to square up the analog signal offloads work from the firmware and guarantees the microcontroller only ever sees clean logic levels, which improves repeatability across temperature and lighting variation.



Figure 7: LM358

3.7 W103 10K Potentiometer / Trimmer

- The W103 is a 10 k Ω trimmer potentiometer used to set an adjustable reference voltage in the conditioning circuit. In Configuration 1 it sets the comparator's threshold, allowing the IR detection point to be tuned to the specific door clearance and ambient lighting of the installation. In Configuration 2 it sets the reference/sensitivity for the LM358B stage, allowing the motion-detection threshold to be calibrated during installation. Because the threshold is adjustable in hardware, the same circuit can be tuned to a wide range of environments without any change to the firmware.

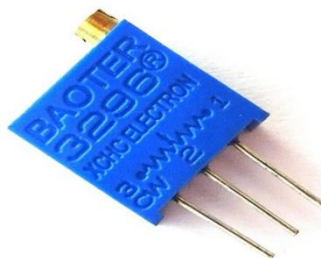


Figure 8: W103 10K

3.8 HC-SR501 PIR Sensor (Configuration 2)

- The HC-SR501 is a passive infrared (PIR) motion sensor and the primary sensing element of Configuration 2. It detects the change in infrared radiation produced by the body heat of a person moving through its wide field of view and outputs a digital HIGH when motion is present. Because its output is already a clean logic level, it connects directly to an ESP32 input pin. The module includes its own on-board potentiometers for adjusting sensitivity (detection range) and output hold time, complementing the W103/LM358B conditioning in the wider circuit. Its low cost, low power consumption, and wide-area coverage make it well-suited to presence detection for both occupied-hours monitoring and after-hours security.



Figure 9: HC-SR501 PIR Sensor

3.9 C945 Transistor (Configuration 2)

- The C945 is an NPN bipolar junction transistor that acts as a switching driver between the ESP32 and the relay coil. The ESP32's GPIO pin can only source a small current, insufficient to energize a relay coil directly. The C945 solves this by using the small GPIO control signal at its base to switch a larger current through its collector–emitter path, supplying the coil current the relay requires. This protects the microcontroller pin from excessive current draw and ensures the relay switches cleanly and reliably on each detection event.

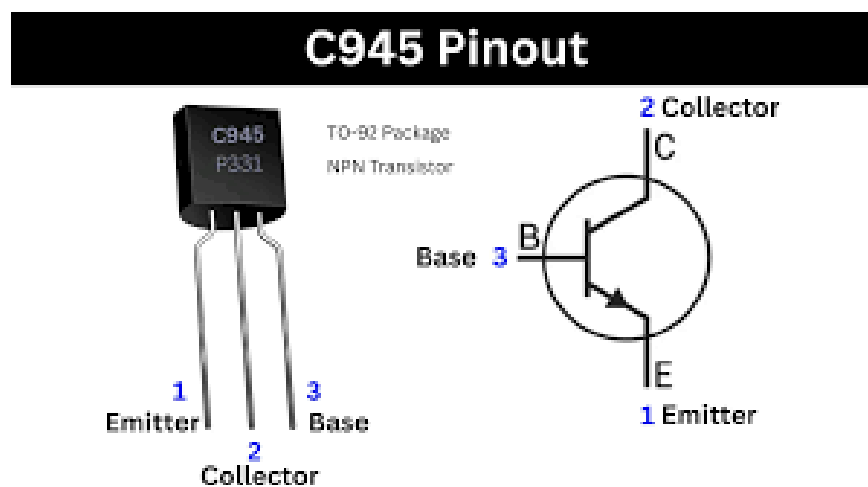


Figure 10: C945 Transistor

3.10 Relay

- The relay is an electromechanical switch that provides electrical isolation between the low-voltage logic and the higher-power output circuit. Its coil is energized by the control side (driven by the ESP32 directly in Configuration 1, or through the C945 transistor in Configuration 2), while its contacts switch a completely separate circuit. In Configuration 1 the relay isolates the buzzer signaling loop; in Configuration 2 the 5 V relay switches the isolated 12 V DC load. This isolation is a core safety feature, preventing switching transients and the higher voltage from reaching the sensitive microcontroller pins.



Figure 11: Relay

3.11 Output Loads: Buzzer and 12V Alerting Lamp/Bell

- The output device differs between the two configurations. In Configuration 1, a buzzer/melody module produces the audible welcome chime on each door-opening event, driven through the relay. In Configuration 2, a 12 V alerting lamp or bell acts as the high-power output load; switched by the relay, it provides a strong, conspicuous presence alert suitable for after-hours monitoring. In both cases the output is electrically separated from the logic by the relay, so the microcontroller controls the alert without ever being directly connected to the signaling or load circuit.



Figure 12: Buzzer and 12V Alerting Lamp/Bell

3.12 Power Supply And Voltage Considerations

- A stable and properly matched power supply is vital to the reliable operation of both configurations. The HW-131 module provides the regulated 3.3 V and 5 V rails that power the ESP32, the LM358B, the sensors, and the relay coil drive. The logic devices share a common 3.3 V/5 V domain and a common ground, while the 12 V load in Configuration 2 is kept in a separate domain that the logic never touches directly, reached only through the relay contacts.
- This separation provides several key benefits: stable sensor readings unaffected by load-switching transients, reliable signal levels presented to the ESP32, and the safe operation of every component. By maintaining voltage compatibility within the logic domain and isolating the high-power domain, the system achieves consistent and dependable performance even during prolonged or continuous operation.

3.13 Hardware Interconnection Strategy

- The hardware components of each configuration are interconnected following a simple yet efficient wiring strategy that ensures reliable operation while keeping circuit complexity to a minimum. In Configuration 1, the IR receiver feeds the LM358B comparator (referenced by the W103 trimmer); the comparator output goes to an ESP32 input, and ESP32 output pins drive the buzzer through the relay. In Configuration 2, the HC-SR501 output goes to an ESP32 input, an LM358B stage with the W103 trimmer conditions the signal where required, and an ESP32 output pin drives the C945 transistor, which in turn energizes the relay coil that switches the isolated 12 V load.
- In both configurations each device receives its appropriate supply voltage and a common ground from the HW-131 module, while the relay forms the single, well-defined crossing point between the logic domain and the higher-power output domain. This minimal, clearly partitioned wiring scheme reduces the likelihood of connection errors and reflects three core design principles, namely simplicity, reliability, and safe isolation, which are particularly important objectives in the development of embedded alert systems.

IV. Software Implementation And Simulation

4.1 ESP32 Board Architecture

- The software side of this project is built on two tools that work together. The first is the Arduino IDE, an open-source development environment created by Arduino, used to write and compile the firmware for the ESP32. The second is the Proteus Design Suite, a circuit design and simulation software developed by Labcenter Electronics (UK), used to simulate the complete circuit and run that firmware on a virtual ESP32 before any physical hardware is built.

- The Arduino IDE was chosen because the ESP32 is fully supported through the Espressif board package, allowing the firmware to be written in the familiar C/C++ Arduino framework with its `setup()` and `loop()` structure and its simple digital I/O functions. In this project, the program essentially performs three tasks: configuring the input/output pins in the initialization stage, reading the logic state from the sensor (through the output of the LM358 comparator), and, based on that state, issuing control commands to the output pin to activate the alert stage. This report does not include the source code or analyse it line by line; instead, it only describes the general idea of the program and how it is loaded into the circuit.

- The implementation process takes place in three steps. In the first step, the firmware is written and compiled (verified) in the Arduino IDE for the ESP32 target. Instead of uploading it directly to a real board, the compilation produces a binary file in the standard `.hex` (or `.bin`) format. This compiled file is the bridge between the two tools. In the second step, within Proteus, we open the properties of the virtual ESP32 component and point it to the path of the newly created `.hex` file, that is, loading the program into the virtual chip just as it would be loaded into a real one. In the third step, when the simulation is run, Proteus executes the firmware exactly as a real chip would: it reads the virtual input pins (the signals

from the sensor and the comparator) and drives the virtual output pins (LED, buzzer, relay) in real time, allowing the behaviour of the whole circuit to be observed directly on the screen.

- Because the firmware running in Proteus is the same binary that would run on the physical ESP32, the observed behaviour reflects the true logic of the program rather than an approximate simulation. The advantage of this approach is that the entire system, both hardware and software, can be tested, debugged, and demonstrated entirely on a computer, which significantly reduces development cost and risk before investing in a physical prototype.

4.2 IR Door-Status Chime Simulation

4.2.1 Picking The Components In Proteus

- The first step in building this circuit is to open Proteus and enter the component selection window (the “Pick Devices” dialog, accessed from the P button in the Devices panel). Each part is added by typing its keyword or part code into the search box and placing it on the schematic. For this configuration the following devices are picked:

- + **ESP32-WROOM:** The microcontroller (central processing unit).
- + **CNY70:** The reflective infrared transmit/receive sensor.
- + **LM358N:** The dual operational amplifier used as the comparator (U1:A).
- + **POT-HG:** The adjustable potentiometer/trimmer used for the threshold (RV1).
- + **RES:** The resistors (R1, R2, R3).
- + **LED-RED:** The visual indicator (D1).
- + **BUZZER:** The audible alert (BUZ1).

→ ***Note:** In this design the CNY70 is used to play the role of the infrared transmit/receive LED pair, because Proteus does not provide a model for that discrete IR LED pair. The CNY70 is a reflective sensor that already*

integrates an infrared emitter and an infrared receiver in a single package, so it reproduces the same transmit/receive behaviour required by the circuit.

4.2.2 Wiring The Components

- With every device placed on the worksheet, the connections are made step by step to form the complete schematic of Configuration 1:
 - + On the CNY70 (Q1), the emitter LED side is connected through R1 (220 Ω) to limit its forward current, and the receiving transistor side is connected through R2 (330 Ω), which acts as a load resistor that converts the transistor current into a usable analog voltage.
 - + This analog voltage node is wired to the non-inverting input of the LM358N (U1:A), while the wiper of RV1 (1 k Ω) is wired to the inverting input to provide the adjustable reference voltage.
 - + The comparator output (pin 1 of U1:A) is wired to an input pin of the ESP32.
 - + An output pin of the ESP32 (D8) is wired through R3 (220 Ω) to the LED-RED (D1), while a separate output pin (D6) drives the BUZZER (BUZ1). The other terminals of the LED and buzzer go to ground.
 - + Finally, the supply and ground rails are connected to all parts so the circuit is powered.

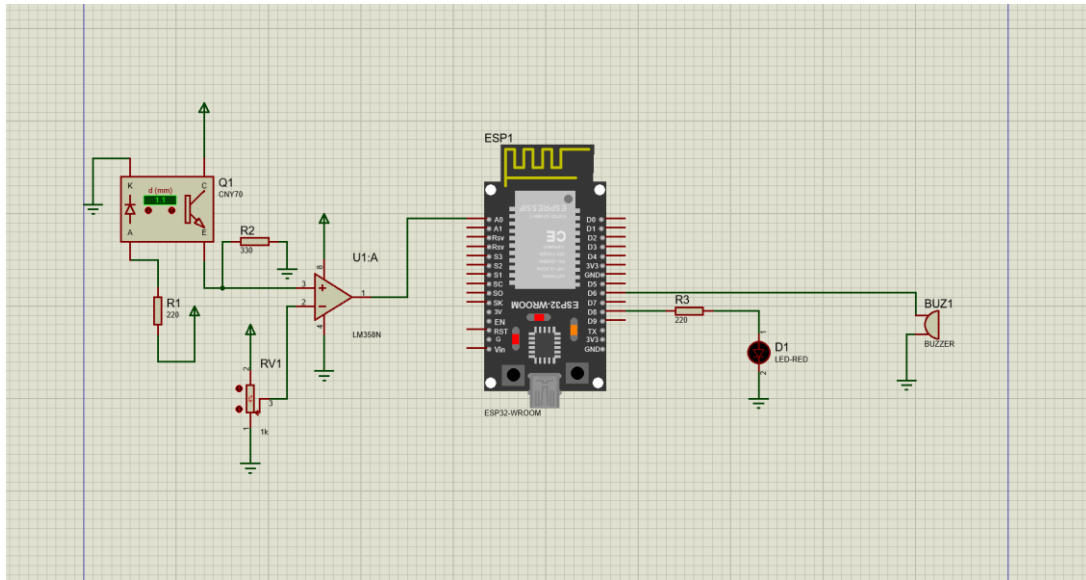


Figure 13: IR door-status chime system in Proteus

4.2.3 How The Circuit Operates

- The behaviour of the circuit is governed by a short Arduino program. In the initialization function (setup()), the firmware configures the pin connected to the comparator output as an input and the pin driving the LED and buzzer as an output, and sets the output LOW so the alert is initially off. In the main loop (loop()), the firmware repeatedly reads the comparator pin with a digital-read function; when that pin is HIGH (a door-opening event), it writes the output pin HIGH to turn on the LED and buzzer, and otherwise writes it LOW to keep them off, with a small delay added for stable, flicker-free operation.
- This program is written and verified in the Arduino IDE, which compiles it into a .hex binary file. In Proteus, the file is loaded by double-clicking the ESP32 component and pasting the path to this .hex file into its “Program File” property. Once the file is attached, pressing the Run button starts the simulation and the ESP32 executes the firmware exactly as a physical chip would.
- With the firmware running, the circuit operates as follows: the CNY70 continuously monitors the door. When the door state changes, the amount of infrared light reflected back to the receiving transistor changes, which alters the analog voltage at the LM358 input. The comparator instantly compares this voltage against the RV1 reference and outputs a clean

digital level to the ESP32. The firmware reads this pin, applies debounce, and, on confirming a door-opening event, sets its output pin HIGH, which lights the LED and sounds the buzzer together.

4.2.4 Operating Principle and Justification

- Detection relies on the comparator switching decisively at a defined threshold. The LM358 in open-loop comparator mode follows the rule:

$$\text{If } V(+)>V(-) \rightarrow \text{output} \approx \text{HIGH}$$

$$\text{If } V(+)<V(-) \rightarrow \text{output} \approx \text{LOW}$$

- Where one input carries the IR-receiver voltage and the other carries the RV1 reference voltage. Because the open-loop gain of the op-amp is very large, even a small difference between the two inputs drives the output fully to one supply rail. This is precisely why the gradual analog change from the infrared sensor becomes a sharp, noise-free digital edge.

- The LED current-limiting resistor R3 is calculated using Ohm's law. Taking the ESP32 output as 3.3 V and a typical red-LED forward voltage of about 2 V:

$$I(LED) = (V(pin) - V(LED)) / R3 = (3.3 - 2.0) / 220 \approx 5.9 \text{ mA}$$

→ This is a safe operating current for both the LED and the ESP32 pin, confirming that the 220 Ω value is appropriate and explaining why the LED lights at normal brightness without overloading the microcontroller.

4.2.5 Simulation Result

- In simulation, adjusting RV1 shifts the switching point of the comparator. When the reflected-light voltage crosses the threshold, the comparator output, the LED, and the buzzer all change state simultaneously, confirming that the circuit behaves exactly as predicted by the comparator rule above.

4.3 PIR Thermal Sensing Notification System

4.3.1 Picking The Components In Proteus

- As with the first configuration, the circuit is built by opening Proteus and entering the “Pick Devices” dialog (the P button in the Devices panel). The following devices are picked for this configuration:

- + **ESP32-WROOM:** The microcontroller (central processing unit).
- + **PIR SENSOR V3:** The passive infrared motion sensor that detects human body heat (PIR1).
- + **LM358N:** The dual operational amplifier used as the comparator/conditioning stage (U1:A).
- + **POT-HG:** The adjustable potentiometer/trimmer used for the threshold (RV1).
- + **2N2222:** The NPN transistor used as the relay driver (Q1).
- + **RES:** The resistor (R1).
- + **RELAY:** The 5 V electromechanical relay that switches the load (RL1).
- + **BATTERY:** The 12 V DC source for the load circuit (BAT1).
- + **BUZZER:** The audible alert load (BUZ1).
- + **LOGICSTATE:** A logic input used during simulation to manually drive/test the PIR signal.

→ **Note:** *In this design the BATTERY (12 V, BAT1) is used to supply the buzzer load, because Proteus does not provide a model for the HW-131 power-supply module used in the physical circuit. The battery reproduces the same regulated DC source that the HW-131 would deliver, so the simulation behaves identically to the real hardware in terms of powering the alert load.*

4.3.2 Wiring The Components

- With every device placed on the worksheet, the connections are made step by step to form the complete schematic of Configuration 2:

- + The output of the PIR SENSOR V3 (PIR1) is wired to the non-inverting input of the LM358N (U1:A), while the wiper of RV1 (10

k Ω) is wired to the inverting input to set the adjustable reference voltage.

- + The comparator output (pin 1 of U1:A) is wired to an input pin of the ESP32 (D4).
- + An output pin of the ESP32 (S0) is wired through R1 (1 k Ω) to the base of the transistor Q1 (2N2222). The emitter of Q1 goes to ground.
- + The collector of Q1 is wired to one end of the relay coil (RL1, 5 V); the other end of the coil goes to the supply, so that energizing the transistor closes the relay.
- + On the contact side, the relay switches the high-power loop formed by the 12 V battery (BAT1) and the buzzer (BUZ1), keeping this loop electrically separate from the logic.
- + Finally, the supply and ground rails are connected to all parts so the circuit is powered.

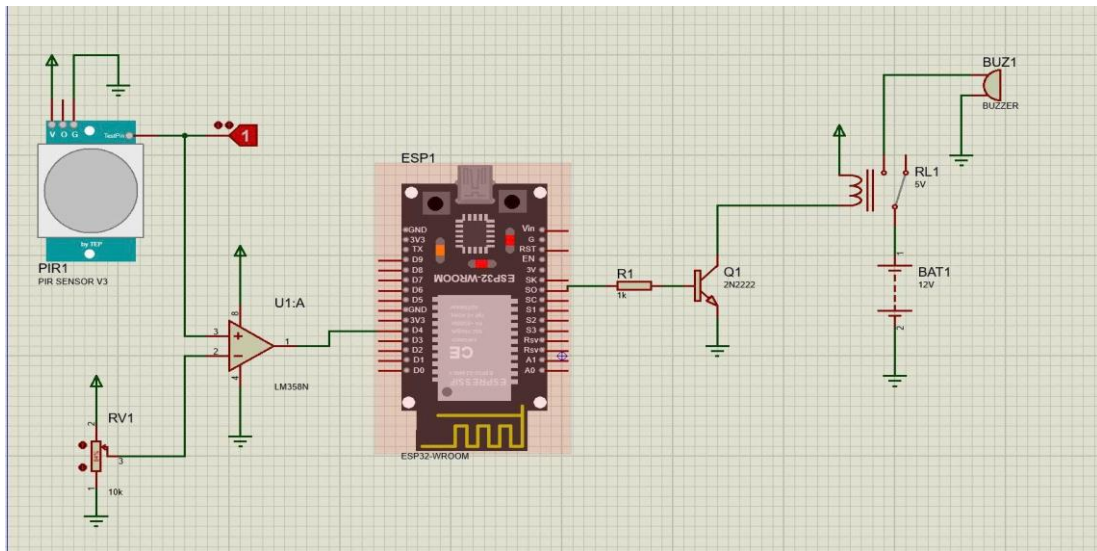


Figure 14: PIR thermal sensing notification system in Proteus

4.3.3 How the Circuit Operates

- The behaviour of the circuit is governed by a short Arduino program. In the initialization function (setup()), the firmware configures the pin connected to the comparator output as an input and the pin driving the transistor as an output, and sets that output LOW so the transistor is off,

the relay is released, and the buzzer is silent. In the main loop (loop()), the firmware repeatedly reads the input pin with a digital-read function; when motion is detected (the pin is HIGH), it writes the output pin HIGH to switch on the transistor and energize the relay, and otherwise writes it LOW to release the relay, with a small delay added for stable operation.

- This program is written and verified in the Arduino IDE, which compiles it into a .hex binary file. In Proteus, the file is loaded by double-clicking the ESP32 component and pasting the path to this .hex file into its “Program File” property. Once the file is attached, pressing the Run button starts the simulation and the ESP32 executes the firmware exactly as a physical chip would.

- With the firmware running, the circuit operates as follows: the PIR SENSOR V3 continuously monitors its field of view. When a person moves within range, the sensor output goes HIGH; this signal passes through the LM358N conditioning stage and reaches the ESP32 input. The firmware reads the pin, confirms the motion event, and sets its output HIGH, which drives the 2N2222 into conduction, energizes the relay coil, and closes the contacts so that the 12 V battery powers the buzzer.

4.3.4 Operating Principle and Justification

- The key design point is using the 2N2222 as a low-side switch for the relay coil. The base resistor R1 sets the base current. With the ESP32 output at 3.3 V and the transistor's base-emitter drop at about 0.7 V:

$$I(base) = (V(pin) - V(BE)) / R1 = (3.3 - 0.7) / 1000 \approx 2.6 \text{ mA}$$

- For a 2N2222 transistor with a typical current gain (β) of about 100, the maximum collector current it can switch is:

$$I(collector) = \beta \times I(base) \approx 100 \times 2.6 \text{ mA} \approx 260 \text{ mA}$$

- A typical 5 V relay coil draws only a few tens of milliamps, far below this 260 mA capability, which proves that the transistor is more than capable of driving the relay and explains why the relay switches reliably whenever the ESP32 pin goes HIGH. This calculation also justifies the

choice of $R1 = 1\text{ k}\Omega$: it provides enough base current to fully saturate the transistor while keeping the ESP32 pin current low and safe.

- The relay itself provides the safety basis for the entire output stage. The microcontroller logic operates at 3.3 V, while the alert load runs from a separate 12 V battery source. These two circuits are never directly connected to each other; the only link is the magnetic coupling inside the relay between its coil and its contacts. This is why a fault on the 12 V side cannot propagate to and damage the ESP32, satisfying the project's safe-load-management objective.

4.3.5 Simulation Result

- In simulation, the PIR signal (driven by the sensor or a LOGICSTATE input for testing) is toggled to represent a person entering the monitored zone. When the signal goes HIGH, the ESP32 output switches the 2N2222 into conduction, the relay closes, and the 12 V battery activates the buzzer; when the signal returns LOW, the relay releases and the buzzer stops. Throughout the test the logic side remains fully isolated from the 12 V load, confirming that the circuit behaves exactly as predicted by the transistor-and-relay analysis above.

4.4 Demonstrability to Customers Through The Product and Simulation

- A key strength of this proposal is that the product's value can be demonstrated directly to customers and investors, rather than only described. This demonstrability is what turns the proposal from a promise into convincing evidence.

- + **Working physical prototype:** both configurations have been built on a breadboard and operate live — the IR chime sounds when the door opens, and the PIR unit triggers the 12V bell when a person is detected. A customer can see and hear the product work in person.

- + **Simulation as proof of reliability:** because the same firmware binary that runs on the real ESP32 is executed on the virtual chip in Proteus, the

simulation is a faithful preview of the hardware. It lets a customer or investor verify that the logic and the circuit behave correctly under different conditions, without needing to build anything.

+ **Two-stage validation builds trust:** by showing the design works first in simulation and then in real hardware, the project gives the customer confidence that the product is complete, reliable, and ready to move toward production with minimal risk.

V. Results, Discussion and Applications

5.1 Introduction

- This chapter presents the results obtained from the simulation and testing of the two configurations developed in this project, namely the IR Door-Status Chime System (Configuration 1) and the PIR Thermal Sensing Notification System (Configuration 2). The objective of the testing phase was to verify that each configuration operated according to its design requirements and achieved the intended detection and alerting functionality. Observations were collected in Proteus, covering sensor response, comparator switching behaviour, transistor and relay operation, and the final output stage (LED, buzzer, and 12V load).

5.2 IR Door-Status Chime Results

- The door-status chime configuration was built physically on a breadboard around an ESP32-WROOM, an infrared transmit/receive LED pair as the sensing element, an LM358B comparator, a W103 10K trimmer, current-limiting resistors, a red LED indicator, and a buzzer, all powered from the HW-131 module. During testing, the door state was changed in front of the IR pair and the response of the LED and buzzer was observed directly.

- **Door closed (red LED off, no chime)** In the standby state, when the door was closed, the infrared signal stayed below the threshold set by the W103 trimmer, so the comparator output remained LOW. The ESP32 read this LOW state and kept both the red LED and the buzzer off, so the circuit stayed silent while powered and waiting. In this state, the system's blue indicator LED blinks slowly to show that the circuit is operating normally and is in standby mode, with no door-opening event detected. This inactive condition is shown in Figure 5.1, where the red LED is off and the buzzer produces no sound.

□

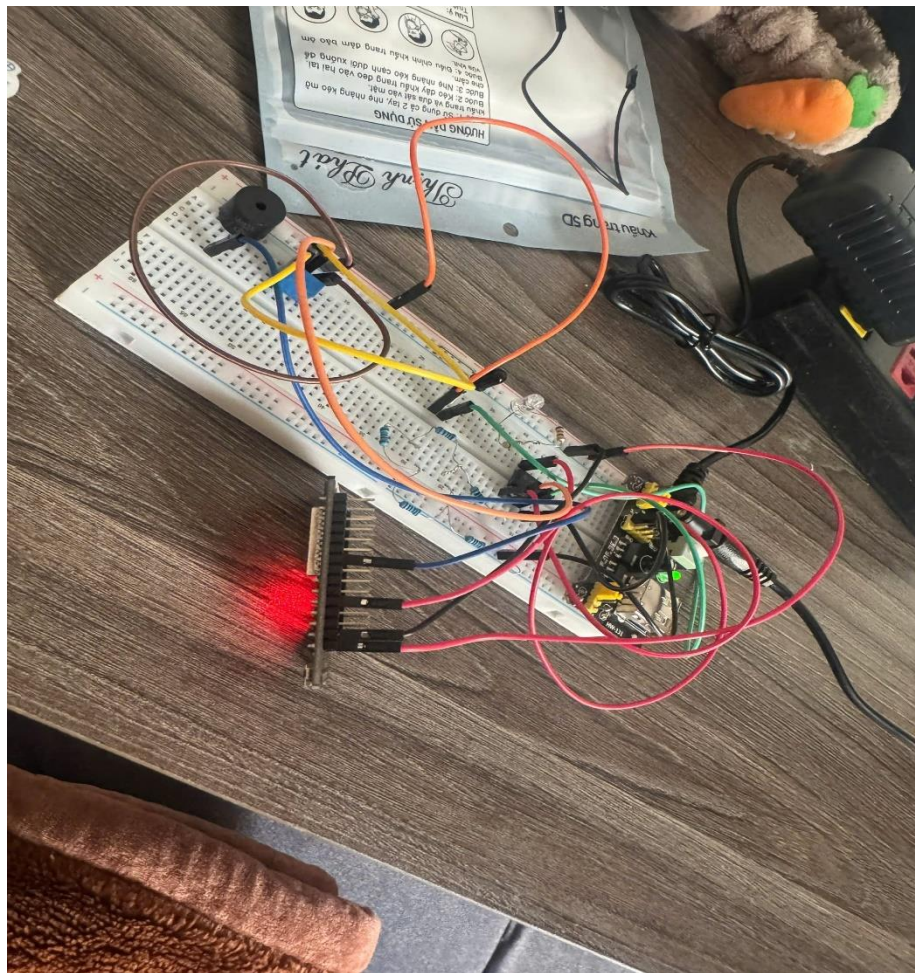


Figure 15: Configuration 1 in the door-closed (standby) state

- **Door open (red LED lit, chime sounding):** When the door was opened, the reflected/interrupted infrared level crossed the threshold; the comparator output switched sharply HIGH, the ESP32 confirmed the

door-opening event, and it drove its output pins HIGH to light the red LED and sound the buzzer together. At this point, the blue indicator LED switches to blinking very fast to show that the system has just detected the door-opening event and is in the alert state. This active condition is shown in Figure 5.2, where the red indicator LED is clearly lit and the buzzer sounds the welcome chime.

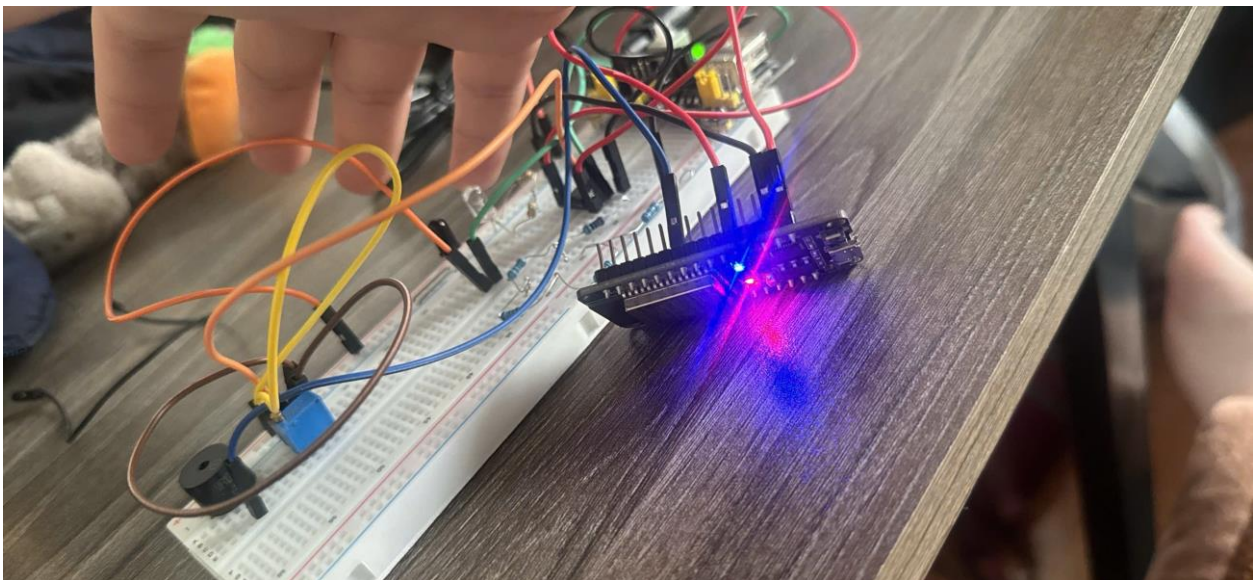


Figure 16: Configuration 1 in the door-open (triggered) state

- The behaviour was verified through direct observation of the red LED and the audible buzzer. The output matched the firmware decision in every trial, and no unstable oscillation occurred at the switching point, confirming that the comparator together with the firmware debounce produced a single clean trigger per door-opening event.

5.3. PIR Thermal Sensing Results

- The PIR thermal sensing configuration was built physically on a breadboard around an ESP32-WROOM, an HC-SR501 PIR sensor as the

sensing element, an LM358B stage with a W103 10K trimmer, a C945 transistor as the relay driver, a 5V relay, and a 12V alerting bell, all powered from the HW-131 module. During testing, a person moved in and out of the sensor's field of view and the response of the relay and the 12V bell was observed directly.

- **No person detected (relay released, bell silent):** In the standby state, when no one was within the sensor's field of view, the HC-SR501 output stayed LOW. The ESP32 read this LOW state and kept its control pin LOW, so the C945 transistor was off, the relay stayed released, and the 12V bell remained silent. In this state, the system's blue indicator LED blinks slowly to show that the circuit is operating normally and is in standby mode, with no presence detected. This inactive condition is shown in Figure 5.3, where the relay is released and the bell produces no sound.

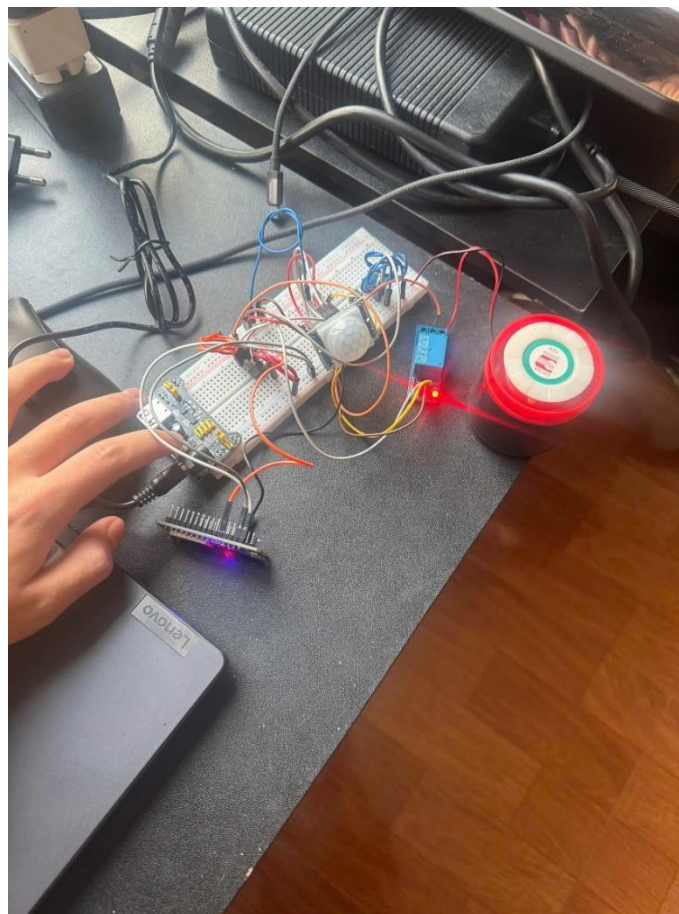


Figure 17: Configuration 2 in the no-person (standby) state

- **Person detected (relay energized, bell sounding):** When a person entered the monitored zone, the HC-SR501 output went HIGH; the ESP32 confirmed the motion event and drove its control pin HIGH, switching the C945 transistor into conduction. The transistor energized the relay coil, the relay contacts closed, and the 12V supply powered the bell to sound the alert. At the same time, the green LED on the relay module lit up to indicate that the relay had switched on, and the ESP32's blue indicator LED switched to blinking very fast to show that the system had just detected human presence and was in the alert state. This active condition is shown in Figure 5.4, where the relay's green LED is lit, the relay indicator LED is on, and the 12V bell sounds the alarm.

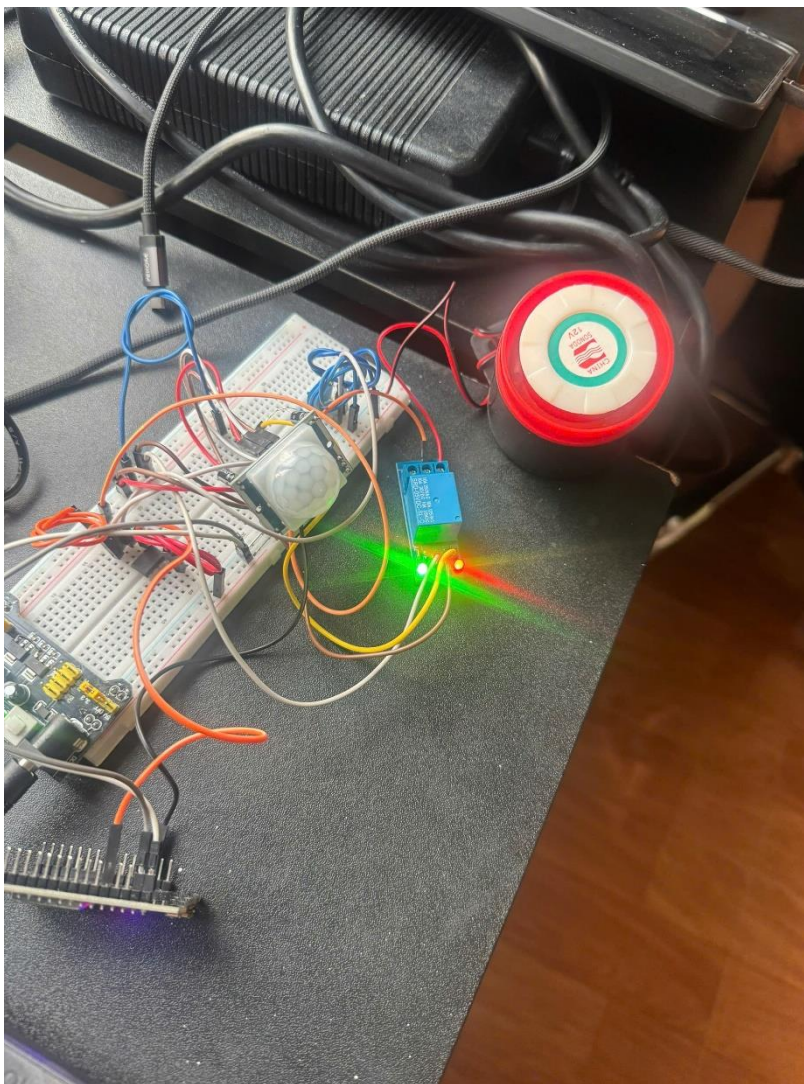


Figure 18: Configuration 2 in the person-detected (triggered) state

Motion Condition	PIR Output	ESP32 Pin	C945	Relay	12VLamp/Bell
No motion	LOW	LOW	OFF	Released	OFF
Person detected	HIGH	HIGH	Conducting	Closed	ON(Alert)
Motion ends	LOW	LOW	OFF	Released	OFF

Table 5.3: Observed operating behaviour of Configuration 2 (PIR thermal sensing)

5.4. Comparator And Threshold Verification

- Additional tests were performed on Configuration 1 to validate the comparator stage independently. The W103 10K trimmer was adjusted across its range while the infrared input was varied, and the switching point of the LM358B comparator was observed together with the red LED and buzzer.
- The comparator consistently changed state when the infrared signal crossed the reference set by the W103 trimmer, confirming that the detection threshold can be tuned in hardware. The red LED and buzzer outputs always matched the comparator decision. These observations verify that the threshold can be calibrated to a specific door clearance and ambient-light condition.

5.5. Comparator And Threshold Verification

- Additional tests were performed on Configuration 2 to validate the transistor-relay drive stage. The calculations in Section 4.3.4 predicted a base current of about 2.6 mA through the 1 k Ω base resistor and a maximum switching capability of about 260 mA for the C945, far above the few tens of milliamps drawn by a 5V relay coil.

In testing, the relay switched reliably every time the ESP32 pin went HIGH, with no failure to pull in or drop out, confirming that the transistor is comfortably saturated and that the 1 k Ω base resistor is an appropriate choice. The green relay-module LED and the 12V bell state matched the ESP32 output decisions in every trial, and the logic side remained fully isolated from the 12V load throughout, validating the safe-load-management design.

5.6. Comparative Evaluation

- After both configurations had been implemented and tested individually, a comparative evaluation was conducted to assess the overall project objectives.

- Configuration 1 used comparator-based threshold detection to recognize a discrete door-opening event and responded with a local audible and visual chime. In contrast, Configuration 2 used PIR motion sensing with a transistor-driven relay to detect wide-area human presence and switch a separate, higher-power 12V alert load. The main characteristics of the two configurations are compared in the table below

Feature	IR Door-Status	PIR Thermal
Controller	ESP32-WROOM	ESP32-WROOM
Sensor	IR transmit/receive LED pair	HC-SR501 PIR sensor
Signal conditioning	LM358B comparator + W103 10K trimmer	LM358B stage + W103 10K trimmer
Output driver	Direct GPIO	C945 transistor
Actuator / load	Red LED + buzzer (low-voltage)	5V relay → 12V bell

Power supply	HW-131	HW-131
Detection type	Discrete door-open event	Wide-area presence / motion
Objective	Automatic entry chime	Presence alerting / security

Table 5.6: Comparison of the main characteristics of the two configurations.

5.7. Applications

- Configuration 1 (IR door-status chime) is suited to retail shops, convenience stores, and hospitality entrances, where an automatic welcome chime announces each customer the moment the door opens, in the familiar convenience-store style, without requiring any button press.
- Configuration 2 (PIR thermal notification) is suited to wide-area presence monitoring such as warehouses, offices, and premises requiring after-hours security, where the relay-switched 12V bell provides a strong, conspicuous alert when a person is detected.
- Because both configurations share the same ESP32 core, HW-131 power module, LM358B conditioning, and relay-based isolation, a manufacturer can build either product, or a combined unit, from one common parts pool and a single firmware base, which lowers the bill of materials and simplifies inventory and maintenance.
- The ESP32's unused Wi-Fi and Bluetooth capability means both configurations can later be extended into connected smart-alert products (remote notification, event logging) with no change to the sensing or output hardware.

VI. Strategy And Competitive Advantages

6.1. Competitive Advantages

- To position the product in the market, it is important to compare it against the alert solutions currently available to local customers. Most existing options fall into three groups, each with clear limitations that this project's design addresses.

+ **Manual push-button doorbells and wired buzzers.** These are the cheapest and most common products on the local market. However, they are entirely manual: a visitor must press a button, so they cannot detect a customer or an intruder automatically. They offer no presence sensing, no after-hours security value, and no upgrade path. Our system removes this manual dependence by detecting the event automatically (a door opening or a person approaching) and responding without any human action.

+ **Standalone PIR alarm modules.** Low-cost PIR motion alarms are widely sold for basic security. While they detect motion, most are single-purpose, drive only a small built-in siren, and connect the load directly to the sensor circuit with little or no electrical isolation. This makes them prone to false triggers and offers no safe way to switch a larger external load. Our Configuration 2 improves on these by adding an LM358B conditioning stage, a tunable threshold, and a transistor-driven relay that safely switches a separate, higher-power 12V load while keeping the logic fully isolated.

+ **Commercial smart-home and CCTV systems.** At the higher end, branded smart sensors and camera systems offer rich features but at a much higher price. They are often expensive to install, depend on a closed ecosystem, and may require a subscription or cloud account to function. For a small shop or warehouse, this is costly and inflexible. Our product delivers the core automatic-alert function at a fraction of the cost, works standalone with no subscription, and still leaves a clear upgrade path to Wi-Fi connectivity when the customer wants it.

- To clearly position our product in the market, it is useful to compare our company's solution directly against the alert products offered by local competitors. While competing solutions tend to focus on a single

function, either a cheap manual doorbell or a basic motion alarm, our product is designed to deliver automatic detection, safe high-power isolation, and two distinct use cases on one shared platform. The following table summarizes how our company compares against these competitors across the criteria that matter most to a buyer, including detection capability, safety, cost, and future upgrade potential:

Criteria	Our Company	Competitors
Detection	Automatic (IR door-status + PIR motion)	Mostly manual or single-sensor
Use cases	Two products on one platform (entry chime + presence alert)	Usually one fixed function
Safety	Relay isolation between logic and 12V load	Little or no isolation
Adjustability	Hardware-tunable threshold (W103 trimmer)	Fixed
Cost	Low (built on inexpensive ESP32)	Cheap but basic, or expensive and complex
Subscription	None, fully standalone	Smart systems often require a subscription
Upgrade path	Built-in Wi-Fi/IoT ready	Usually none
Validation	Proven in simulation + real prototype	Often not demonstrated

Table 6.1: Comparison of our company and other local competitors.

6.2. Business Model

- To ensure sustainable development and scalability, the Motion-Activated Electronic Notification System is built on the Business Model Canvas (BMC) framework. This model clearly identifies the target customers, core value proposition, revenue streams, key resources, operational activities, and cost structure of the business. By combining low-cost embedded hardware with the growing demand for affordable, automatic security and entry-alert solutions, the project aims to build a scalable family of alert products rather than a single one-off device.

Motion-Activated Electronic Notification System — Business Model Canvas

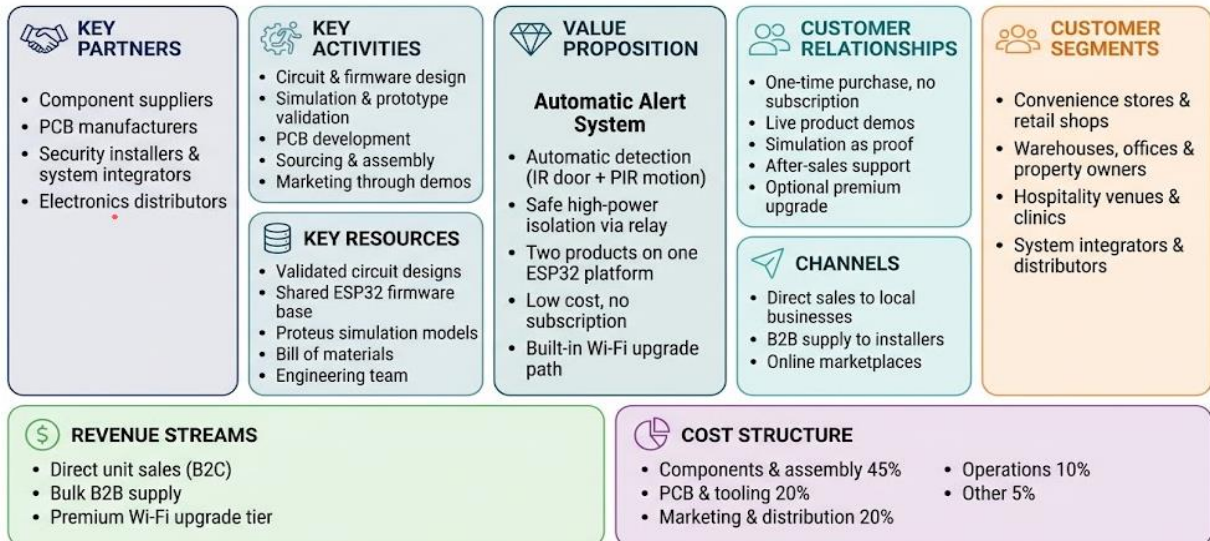


Figure 19: Business Model Canvas of the Motion-Activated Alert System

- The **target customers** include convenience stores and retail shops seeking an automatic entry chime, warehouses, offices and property owners needing wide-area presence and after-hours security, hospitality venues and clinics wanting a discreet visitor tone, and system integrators or distributors who resell or bundle the product. These customer groups share a clear need for affordable, automatic alerting and are willing to adopt a low-cost standalone device.

- The **core value proposition** lies in delivering automatic detection, safe high-power isolation, and two distinct use cases (entry chime and presence alarm) on a single shared ESP32 platform, at a fraction of the cost of commercial smart-home or CCTV systems, with no subscription and a built-in upgrade path to Wi-Fi. This combination of low cost, safety, and flexibility creates a clear differentiation from existing products in the market.

- To reach customers, the product is distributed through **channels** including direct sales to local businesses, B2B supply to security installers and electronics distributors, and online marketplaces. Partnerships with installers and integrators help increase reach and expand the potential customer base.

- Regarding **customer relationships**, the product is sold as a standalone one-time purchase with no subscription, supported by live product demos and simulation evidence to build buyer confidence, plus after-

sales support and an optional premium Wi-Fi upgrade to retain customers over time.

- The **revenue model** is designed around diversification: direct unit sales (B2C), bulk B2B supply, and a premium connected (Wi-Fi) upgrade tier. A diversified revenue structure helps the business minimize risk and enhance its growth potential.

- The **key resources** include the validated circuit designs for both configurations, the shared ESP32 firmware base, the Proteus simulation models, the bill of materials, and the engineering team. These strategic assets form the foundation of the product's competitive advantage. The company's **key activities** include circuit and firmware design, simulation and prototype validation, PCB development, sourcing and assembly, and marketing through demos and trade channels, together with continuous improvement based on field testing.

- The **key partners** include component suppliers, PCB manufacturers, security installers and system integrators, and electronics distributors. This partner ecosystem supports production and market expansion. Regarding the **cost structure**, the largest share is the per-unit component and assembly cost, followed by one-time PCB design and tooling, then marketing and distribution, with minimal overhead since the base product is standalone and requires no servers.

Component	Description
Customer Segments	Convenience stores, retail shops, warehouses, offices, clinics, integrators.
Value Proposition	Automatic detection, safe isolation, two configurations, low cost, Wi-Fi upgrade path.
Channels	Direct sales, B2B supply, online marketplaces.
Customer Relationships	One-time purchase, no subscription, demos, after-sales support, premium upgrade.
Revenue Streams	Direct unit sales, bulk B2B, premium Wi-Fi upgrade tier.
Key Resources	Validated circuit designs, shared firmware, simulation models, BOM, engineering team.
Key Activities	Design, simulation & prototype validation, PCB development, assembly, marketing.
Key Partners	Component suppliers, PCB manufacturers, installers, distributors.
Cost Structure	Component & assembly cost, PCB/tooling, marketing & distribution, minimal overhead.

Table 6.2: Business Model Canvas of the Motion-Activated Alert System.

6.3 Pricing Strategy (Cost And Selling Price)

- The pricing strategy is built on the product's low production cost. Because the system is built entirely from inexpensive, widely available components, it can be sold at a price that is easy for a small business to afford while still returning a profit margin, especially once parts are bought in bulk for volume production.

+ **Production cost:** built around an affordable ESP32 platform and standard supporting components, the estimated material cost is only about 110,000 VND per unit for Configuration 1 (IR door-status chime)

and about 140,000 VND per unit for Configuration 2 (PIR thermal notification) at volume pricing.

+ **Customer-friendly selling price:** Configuration 1 is priced at around 249,000 VND and Configuration 2 at around 299,000 VND. These are affordable, easy-to-decide prices for a small shop or warehouse owner, far below the cost of commercial smart-home or CCTV systems.

+ **Positioning:** the product is deliberately priced as a low-cost, standalone alternative to expensive smart-home systems, which often cost several million VND and require a subscription, while still offering far more automation and safety than a basic doorbell or PIR alarm.

6.4 Marketing Plan (4P Strategy)

- The marketing plan of the Motion-Activated Electronic Notification System is developed based on the Marketing Mix 4P framework, which consists of Product, Price, Place, and Promotion. This model is suitable for analyzing how the company introduces its product to the market, positions its value proposition, selects distribution channels, and develops communication strategies to reach target customers. For a low-cost hardware product such as this alert system, the marketing strategy is intended not only to drive initial sales but also to build trust in the product's reliability, reach both individual buyers and bulk B2B partners, and establish the brand as an affordable, dependable choice in the security and entry-alert market.

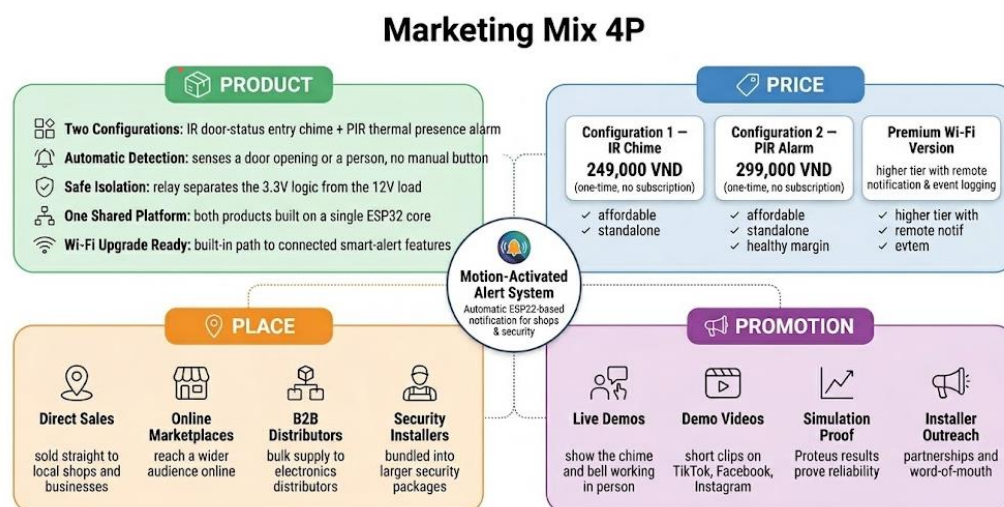


Figure 20: Marketing Mix 4P Strategy of the Motion-Activated Alert System

- Regarding the **Product** element, the system is positioned as a low-cost, automatic alert device offered in two configurations on a single shared ESP32 platform: an IR door-status entry chime (Configuration 1) and a PIR thermal presence alarm (Configuration 2). The product focuses on solving key problems faced by small businesses, including the need for automatic detection without a manual button, safe switching of higher-power alerting loads, and an affordable alternative to costly smart-home systems. Rather than locking the customer into a closed ecosystem, the product works standalone out of the box and offers a clear upgrade path to Wi-Fi connectivity, making it both a simple entry-level product and a platform for future smart-alert features.

- Regarding the **Price** element, the product adopts an affordable, one-time-purchase pricing model to reduce barriers for small businesses. Configuration 1 is priced at around 249,000 VND and Configuration 2 at around 299,000 VND, with no subscription required. This pricing is designed to align with the purchasing power of small shops and premises owners while keeping a healthy margin thanks to the low production cost. A premium Wi-Fi-enabled version can be offered at a higher price point for customers who want remote notification, allowing the product to serve different customer segments from basic standalone buyers to those seeking connected features.

- Regarding the **Place** element, the product prioritizes a mix of direct and partner-based distribution channels. It is sold directly to local businesses such as convenience stores, shops, and warehouses, and through online marketplaces to reach a wider audience. In addition, the company develops B2B distribution through partnerships with security installers, system integrators, and electronics distributors, who can resell the device or bundle it into larger security packages. This combination of direct sales, online reach, and B2B partnerships increases credibility, creates direct customer touchpoints, and supports bulk-order volume.

- Regarding the **Promotion** element, the product focuses on practical, demonstration-driven communication. Because the value of an alert device is best shown rather than described, marketing emphasizes live product demos and short videos showing the system in action (the door chime sounding, the PIR triggering the 12V bell), supported by the simulation evidence that proves reliability. These are promoted through

social media channels such as TikTok, Facebook, and Instagram, as well as targeted outreach to small-business owners and security installers. In addition to organic content, partnerships with local trade channels and word-of-mouth from early adopters help build trust and drive adoption cost-effectively.

Element	Strategy	Implementation
Product	Low-cost automatic alert device, two configurations on one ESP32 platform	IR door chime + PIR presence alarm; safe relay isolation; standalone; Wi-Fi upgrade path
Price	Affordable one-time purchase, no subscription	Config 1 $\approx$ 249,000 VND; Config 2 $\approx$ 299,000 VND; premium Wi-Fi version at higher price
Place	Direct + partner-based distribution	Direct sales to businesses, online marketplaces, B2B via installers and distributors
Promotion	Demonstration-driven and social marketing	Live demos and demo videos, simulation proof, TikTok/Facebook/Instagram, installer outreach

Table 6.4: Marketing Mix 4P Strategy of the Motion-Activated Alert System.

6.5 Project Timeline and Milestones

The project follows a clear, time-bound plan from initial design to a deployable product, demonstrating to an investor that the project is well-organized and can be delivered on schedule. The full task breakdown and assignments are given in the Project Progress Timeline; the key milestones are:

- + Milestone 1 — Design & simulation (completed): both configurations designed and fully validated in Proteus.
- + Milestone 2 — Physical prototype (completed): both circuits built and tested on a breadboard, confirming real-world operation.
- + Milestone 3 — PCB development (proposed): replace the breadboard with a purpose-designed PCB for durability and manufacturing.

- + Milestone 4 — Pilot production & field testing (proposed): small manufacturing batch and on-site testing with target customers.
- + Milestone 5 — Connected version (proposed): add Wi-Fi remote notification as a premium product tier.

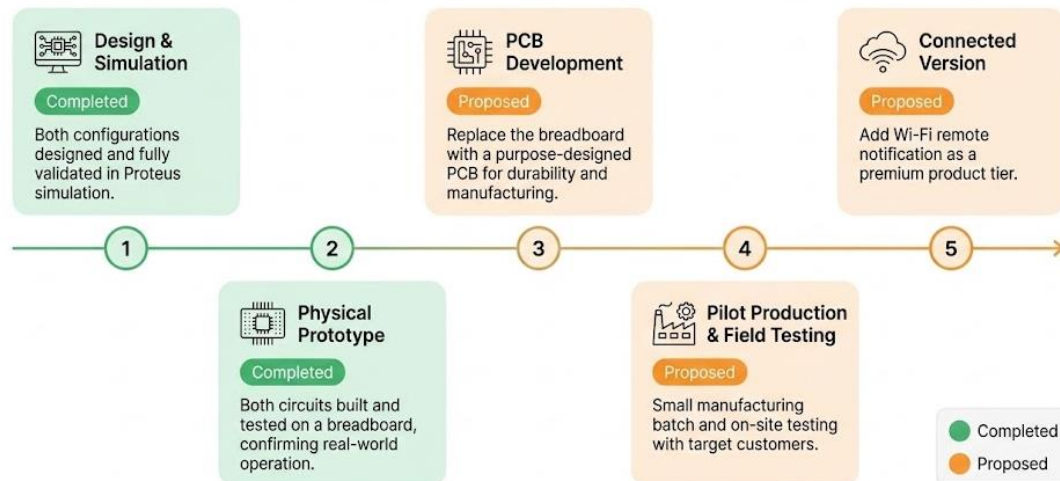


Figure 21: Project Roadmap & Milestones

6.6 Financial Plan and Feasibility

- The financial plan shows that the project is feasible to launch with modest capital and can reach profitability quickly thanks to its low unit cost.
- + Initial capital: covers PCB design, a pilot production batch, basic tooling, and initial marketing — a relatively small upfront investment because the design is already validated.
- + Operating costs: mainly per-unit component and assembly cost, which scales directly with production volume, plus minimal overhead since the product is standalone (no servers required for the base version).
- + Revenue forecast: driven by unit sales across the three channels (direct, B2B, premium upgrade), with margin improving as volume grows and the premium tier is introduced.
- + Break-even: because each unit sells above its low production cost, the business reaches break-even after a modest number of units, making the risk profile attractive.

6.7 Why This Project Is Worth Investing

- For a prospective investor or business partner, the Motion-Activated Electronic Notification System represents a low-risk, high-potential opportunity. Unlike many early-stage proposals that rely on unproven concepts, this project has already been validated in both simulation and physical hardware, meaning the major technical uncertainty has been removed before any production capital is committed. The following points summarize why this project is worth investing in:

+ **Proven and validated technology.** The design works in both Proteus simulation and a real breadboard prototype, with measured behaviour matching the theoretical predictions. Because the same firmware that runs in simulation runs on the physical chip, an investor can be confident that the product performs as claimed and that the remaining step to manufacturing carries minimal technical risk.

+ **Low production cost, strong margin.** The system is built entirely from inexpensive, widely available components on the affordable ESP32 platform. With a volume production cost of only around 110,000–140,000 VND per unit and a customer-friendly selling price of 249,000–299,000 VND, each unit returns a gross margin of roughly 53–55%, which improves further as order volume grows. This means the business can offer attractive prices to customers while remaining profitable on every sale.

+ **Scalable shared-platform design.** Both configurations are built on one common ESP32 core and a single firmware base. This shared-platform strategy lets a manufacturer produce either product, or a combined unit, from one parts pool, lowering the bill of materials, simplifying inventory and assembly, and reducing the cost and complexity of scaling up production.

+ **Clear market need and multiple sales channels.** The product addresses a real, unmet need for affordable, automatic alerting among

convenience stores, retail shops, warehouses, offices, and clinics. Revenue is not dependent on a single channel: the product can be sold directly to businesses, in bulk to B2B partners and installers, and online, which diversifies income and reduces market risk.

+ Built-in growth path and recurring revenue potential. The ESP32's unused Wi-Fi and Bluetooth capability allows the same hardware to be upgraded into a connected smart-alert product, offering remote notification and event logging, with no hardware redesign. This opens a higher-margin premium tier and a potential recurring-revenue stream, giving the business clear room to grow beyond the initial product.

+ Modest capital requirement, fast path to return. Because the design is already validated, the required investment is relatively small, covering PCB design, a pilot production batch, basic tooling, and initial marketing. With a low unit cost and a healthy margin, the business can reach break-even after a modest number of units sold, making the time to return on investment short.

In summary, the investment funds a well-defined step, taking a fully validated prototype into pilot production and the market. The combination of proven technology, low cost, a scalable platform, a clear market, and a built-in upgrade path makes this a low-risk opportunity with strong potential for a profitable, growing product family.

VII. Conclusion And Future Work

7.1 Conclusion

- This project has presented the complete design, simulation, and prototype testing of a Motion-Activated Electronic Alert System built on the ESP32 platform and delivered as two independent configurations.
- Configuration 1, the IR Door-Status Chime System, uses an infrared transmit/receive LED pair and an LM358B comparator, with its

threshold calibrated by a W103 10K trimmer, to detect a door-opening event and respond immediately with a relay-isolated buzzer chime and an LED indicator.

- Configuration 2, the PIR Thermal Sensing Notification System, uses an HC-SR501 sensor to detect human presence and drives a C945 transistor that switches a 5V relay, safely energizing an isolated 12V alerting bell while keeping the high-power load fully separated from the logic.
- Both configurations were first verified in Proteus, where the same firmware that runs on the physical chip was executed on a virtual ESP32, and then confirmed on a physical breadboard prototype. In every test the observed behaviour matched the theoretical predictions and the supporting calculations, including the comparator switching rule and the transistor base and collector current analysis. The results demonstrate that the system meets its central objective: an automatic, low-latency, and electrically isolated response to either a door-opening event or human presence, built from inexpensive and widely available components.
- Beyond the technical results, the project also achieved its broader goals. By validating the complete design in simulation before committing to hardware, development cost and risk were kept low, and the shared-platform strategy, in which both configurations reuse the same ESP32 core, HW-131 power module, LM358B conditioning, and relay-based isolation, demonstrates clear commercial feasibility for a manufacturer building a small family of low-cost alert products.

7.2 Risk Management

- Any product moving from prototype to market carries risks. Identifying these risks early and presenting a clear mitigation plan demonstrates to an investor that the project is realistic and well-managed. The main risks for the Motion-Activated Electronic Notification System fall into technical, market, supply, and financial categories.

- + **Technical risk:** the prototype may behave differently from the breadboard build once moved to a PCB, or may face interference (for example, ambient infrared affecting the IR sensor). This risk is already low because the design has been validated in both simulation and hardware, and it is further reduced by the hardware-tunable threshold (W103 trimmer) and the planned PCB testing stage.
- + **Market risk:** customers may be slow to adopt the product, or competitors may offer cheaper alternatives. This is mitigated by the deliberately low, affordable price, the demonstration-driven marketing that proves reliability, and the two-configuration range that lets the product serve several customer segments rather than depending on one.
- + **Supply chain risk:** component shortages or price increases (especially for the ESP32) could affect production cost. This is mitigated by using common, widely available parts with multiple suppliers, and by the shared-platform design that keeps the bill of materials small and easy to source in bulk.
- + **Financial risk:** the pilot production batch may not sell as quickly as expected, slowing the return on investment. This is mitigated by the low capital requirement, the healthy per-unit margin, and the diversified sales channels (direct, B2B, and online), which spread the risk across several revenue streams.
- + **Safety and quality risk:** an electrical fault on the high-power side could damage the product or harm the user. This is directly addressed by the core design: the relay provides complete electrical isolation between the 3.3V logic and the 12V load, so a high-power fault cannot reach the microcontroller.

7.3 Future Work

- Although the system met its objectives, several directions could extend and strengthen it in future work:
- PCB implementation: replace the breadboard prototype with a purpose-designed printed circuit board to improve mechanical durability, electrical stability, and suitability for mass production.

- **Remote notification:** enable the ESP32's built-in Wi-Fi to send alerts to a phone or cloud dashboard, turning either configuration into a connected smart-alert device with no change to the sensing or output hardware.
- **Event logging:** record door-opening and presence events with timestamps to support later review of entrance patterns and after-hours activity.
- **Adaptive threshold tuning:** add automatic calibration so the detection threshold adjusts to ambient conditions, reducing the need for manual adjustment of the W103 trimmer.
- **Combined unit:** merge both configurations onto a single board so one product can provide automatic entry chiming and wide-area presence alerting together.
- These improvements would build directly on the validated design presented in this report and move the system from a working prototype toward a deployable commercial product.

VIII. Project Progress Timeline

No.	Planned date	Circuit-build task / Activity	Deliverable / Output	Assignee(s)
1	14/05/2026	Conduct a team meeting to align on alert-system requirements; divide into Configuration 1 (IR door-status) and Configuration 2 (PIR thermal) subsystems; list precise components to purchase.	Project modification outline v2.	Whole team
2	15/05/2026	Conduct price research and purchase hardware components for Configuration 1 (IR transmit/receive LED pair, LM358B op-amp, W103 10K trimmer, relay, buzzer/melody module, ESP32, HW-131 supply).	Complete Config 1 hardware inventory list.	Nguyen Bao Khanh
3	16/05/2026	Conduct price research and purchase hardware components for Configuration 2 (HC-SR501 PIR sensor, LM358B op-amp, W103 10K trimmer, C945 transistor, 5V relay, 12V alerting lamp/bell, ESP32, HW-131 supply).	Complete Config 2 hardware inventory list.	Pham Hai Nam
4	18/05/2026	Use a digital multimeter and the HW-131 supply to test and verify the IR transmit/receive LED pair, LM358B, relay, buzzer, and ESP32 board.	Initial Config 1 components test checklist report.	Nguyen Bao Khanh
5	18/05/2026	Use a digital multimeter to test and verify the HC-SR501 sensor, C945 transistor, 5V relay, 12V lamp/bell, and ESP32 board.	Initial Config 2 components test checklist report.	Pham Hai Nam
6	19/05/2026	Create a new project workspace in Proteus, set up the general schematic design frame, and integrate the required component libraries.	Initial Proteus project file configuration.	Whole team
7	21/05/2026	Design the schematic connection lines for Configuration 1 (IR pair to LM358B comparator with W103 trimmer to ESP32 to relay to buzzer) inside Proteus.	Virtual Config 1 schematic drawing file.	Nguyen Bao Khanh
8	21/05/2026	Design the schematic connection lines for Configuration 2 (HC-SR501 to LM358B stage with W103	Virtual Config 2 schematic drawing file.	Pham Hai Nam

No.	Planned date	Circuit-build task / Activity	Deliverable / Output	Assignee(s)
		trimmer to ESP32 to C945 to 5V relay to 12V lamp/bell) inside Proteus.		
9	22/05/2026	Wire and connect the physical Configuration 1 hardware on a breadboard, powered by the HW-131 module, to prepare for firmware testing.	Assembled Config 1 on breadboard.	Nguyen Bao Khanh
10	22/05/2026	Wire and connect the physical Configuration 2 hardware (C945 driver, 5V relay, 12V load) on a breadboard.	Assembled Config 2 on breadboard.	Pham Hai Nam
11	24/05/2026	Perform permanent wiring and tune the W103 10K trimmer of the IR comparator front-end for stable door-status detection.	Calibrated Config 1 sensing circuit.	Nguyen Bao Khanh
12	24/05/2026	Perform permanent wiring and soldering of the C945 transistor relay driver, ensuring secure isolation between the 3.3V logic and the 12V load.	Config 2 relay-driver circuit integration.	Pham Hai Nam
13	26/05/2026	Program the source code to read the LM358B comparator signal and drive the buzzer/melody through the relay on a door-opening event.	Config 1 firmware package.	Nguyen Bao Khanh
14	26/05/2026	Program the source code for HC-SR501 motion detection and automatic C945/relay switching of the 12V alert load.	Config 2 firmware package.	Pham Hai Nam
15	27/05/2026	Link the compiled .hex files into the Proteus environment to simulate and optimize the execution logic for both configurations.	Verified simulation model operation.	Whole team
16	28/05/2026	Tune detection thresholds (W103 10K trimmers and the HC-SR501 on-board pots) and debounce timing; run functional test cases for both configurations.	Tuned parameters and functional test report.	Whole team
17	29/05/2026	Optimize all physical wiring paths and secure each configuration inside	Two finalized standalone product assemblies.	Whole team

No.	Planned date	Circuit-build task / Activity	Deliverable / Output	Assignee(s)
		an independent engineering enclosure for protection.		
18	30/05/2026	Set up test environments for each configuration (door open/close for Config 1, motion for Config 2) to film the product video demo.	High-quality raw video footage for demonstration.	Whole team
19	01/06/2026	Collect technical descriptions, incorporate Proteus schematics, detail the state-machine flow, and compile the final project report.	Final project report manuscript (.docx file).	Whole team
20	02/06/2026	Complete video post-production (cutting, audio balancing, subtitling), design presentation slides, and submit the unified project package online.	Final project submission package (Circuit + Report + Video).	Whole team
21	11/06/2026	Restructure the report into a proper engineering proposal: add the Project Question (Section 1.3), Target Customers and Stakeholders, and Design Basis and Related Work (survey of existing projects, design rationale, and published references) to recover the lost grading criteria.	Updated proposal sections with project question, stakeholders, and referenced design basis.	Whole team
22	11/06/2026	Add the Novelty and Contribution section and a dedicated Business chapter (Section 6): Business Model Canvas, Pricing Strategy, Marketing Mix 4P, Project Roadmap & Milestones, Financial Plan, Investment Opportunity, and Risk Management.	Completed business and investment chapter for the proposal.	Whole team

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